

OEM TECHNICAL OPERATING MANUAL: MODEL CP-194 PROCESS PUMP

1. Technical Specifications

Manufacturer: FlowServe Industrial Pumps Ltd
Model: CP-194 Heavy-Duty Single-Stage Process Centrifugal Pump
Rated Volumetric Flow: 140 m3/h (Normal operating range: 110 - 165 m3/h)
Rated Total Differential Head: 92 meters
Operating Rotational Speed: 2950 RPM
Electric Driver: 55 kW 415V 50Hz 3-phase electric motor
Impeller Diameter: 245 mm closed dynamic balanced impeller

2. Bearing Assembly & Lubrication Limits

Drive-End Bearing: SKF-6314-2Z Deep Groove Ball Bearing
Non-Drive-End Bearing: SKF-NU-314 Cylindrical Roller Bearing
Recommended Lubricant: ISO VG 46 high-demulsibility mineral turbine oil
Maximum Continuous Operating Temperature: 85.0 deg C
High Temperature Alarm: 90.0 deg C | High-High Emergency Trip: 95.0 deg C
Vibration Baseline Norm: ISO 10816-3 Class II (Rigid Foundation: Alarm 4.5 mm/s, Trip 7.1 mm/s)

3. Mechanical Seal Operation

Cartridge Mechanical Seal: BURG-M7N balanced single seal with plan 11 flush piping.
Seal face materials: Silicon Carbide vs Reaction Bonded Tungsten Carbide.